

NWIS OFFICIAL GEOLOGICAL TECHNICAL REPORT

Report Title: **Formation & Geological Summary** | Report ID: **REP-20260929-3601**
Well: **Duliajan-92 (DUL_92)** | Field: **Duliajan Field**

Report Property	Value	Report Property	Value
Generated By:	Authorized Geologist (GEOLOGIST)	Reporting Period:	Last 24 Hours
Generated At:	2026-09-29T11:45:29Z	Depth Interval:	2100.0m - 2210.0m
Provenance Hash:	PROV-GEO-2026-74001F	Status:	COMPLETED

EXECUTIVE GEOLOGICAL SUMMARY

Geological Summary for Duliajan-92: Well currently penetrates Barail Group Sandstone at 2210.0m. Wireline gamma ray and density logs correlate with offset well Duliajan-88 with 94.2% structural similarity.

1. FORMATION & GEOLOGICAL SUMMARY

Formation Target: Barail Group Shale / Sandstone

Lithology Description: Interbedded Fine-to-Coarse Quartzose Sandstone, Siltstone, & Hydratable Smectite Shale

Log Observations: Pronounced Gamma Ray deflection at 2,208m depth MD marks the lithological interface into high-smectite hydratable shale.

Geological Interpretation: Deltaic transgressive phase deposit. Reactive clays require KCl/Poly-Glycol mud system to inhibit swelling.

2. HISTORICAL INCIDENT & EVENT SUMMARY

Historical Well: Duliajan-88 | **Event Type:** Stuck Pipe / Reactive Shale Swelling

Event Depth: 2210.0m | **NPT Duration:** 14.5 Hours NPT

Description: Drillstring experienced severe torque spike from 18.5 kN·m to 79.6 kN·m with 45 tonnes overpull during wiper trip at 2,210m MD.

Context & Evidence: In-situ stress release and osmotic water absorption in smectite clay under low mud weight (1.16 g/cc EMW).

3. GEOLOGICAL RESEARCH REPORT

Research Title: Hazard Assessment & Clay Hydration Dynamics in Barail Group Shales

Findings: Mud weight must be elevated to 1.25 g/cc EMW with 4.0% Poly-Glycol prior to penetrating 2,200m MD.

Interpretation: Overpressure transition zone starts at 2,205m; narrow mud weight window (1.20 – 1.32 g/cc EMW).

4. FORMATION CORRELATION REPORT

Selected Correlation Wells: DUL_92 (Target), DUL_88 (Offset 1.2km), NHK_45 (Offset 3.8km)

Lithology Comparison: DUL_92 displays higher sand-to-shale ratio (0.62) vs NHK_45 (0.48) in Barail interval.

Structural Dip & Correlation: Continuous reservoir sand body with uniform pressure regime; shale hydration risk is field-wide.

DRILLING TELEMETRY & LOG VALUES

Parameter	Value	Unit	Min	Max	Average
Depth (MD)	2210.0	m	0.0	3500.0	2210.0
Rate of Penetration (ROP)	10.7	m/hr	4.0	28.5	10.66
Weight on Bit (WOB)	28.9	kN	10.0	160.0	28.93

Rotary Speed (RPM)	117	rpm	60.0	150.0	117.04
Surface Torque	29.5	kN·m	8.0	85.0	29.53
Standpipe Pressure (SPP)	3374.6	psi	1100.0	3200.0	3374.58
Flow Rate In	2450	L/min	1200.0	2800.0	2450.0
Hook Load	166	kN	800.0	1650.0	166.38